

Thermodynamics (HL)

Everything revolves around one accounting equation: $Q = \Delta U + W$. The process tells you which term becomes zero or changes.

SYSTEMS & SIGN CONVENTION

Closed vs isolated

- Closed: no mass transfer; energy may cross as heat/work.
- Isolated: no mass transfer and no energy transfer.

Clausius signs

- $Q +$: heat enters system.
- $\Delta U +$: internal energy increases.
- $W +$: system does work on surroundings.
- $W -$: surroundings do work on system.

THE CORE FORMULAS

$$Q = \Delta U + W$$

$$W = P\Delta V \cdot \Delta U = \frac{3}{2} nR\Delta T$$

P-V memory: work done = area under the P-V path. Expansion $\rightarrow$ W positive; compression $\rightarrow$ W negative.

FOUR PROCESSES — MEMORIZE THIS TABLE

Process	Constant	Key result	Typical expansion signs
Isochoric	V	$W = 0$; $Q = \Delta U$	not applicable: $\Delta V = 0$
Isobaric	P	$W = P\Delta V$	$\Delta U +$, $W +$, $Q +$
Isothermal	T	$\Delta U = 0$; $Q = W$	$\Delta U 0$, $W +$, $Q +$
Adiabatic	$Q = 0$	$\Delta U = -W$	$\Delta U -$, $W +$, $Q 0$

HEAT ENGINES & CARNOT

Heat engine

- Cycle returns to starting state $\rightarrow \Delta U = 0$ over one complete cycle.
- $\eta = (Q_H - Q_C)/Q_H$.
- Useful work = heat in – heat rejected.

Carnot cycle

- Four reversible stages: isothermal expansion $\rightarrow$ adiabatic expansion $\rightarrow$ isothermal compression $\rightarrow$ adiabatic compression.
- Maximum possible efficiency between two reservoirs.
- $\eta = 1 - T_C/T_H$.

ENTROPY

Second law

- In an isolated system, total entropy increases during spontaneous processes.
- $\Delta S = \Delta Q/T$ for the stated macroscopic relation.
- $S = k_B \ln \Omega$ microscopically.

Exam memory

- Adiabatic reversible process: $\Delta Q = 0 \rightarrow$ entropy remains constant in the material's stated treatment.
- Entropy is a state quantity; a complete cycle returns ΔS of the system to zero.

TOP EXAM TRAPS

Do not mix these up

- Isothermal $\neq$ adiabatic: isothermal means T constant; adiabatic means $Q = 0$.
- W is not always $P\Delta V$ for a changing-pressure path; P-V area is the safe interpretation.
- Carnot temperatures must be Kelvin.
- Over a complete cycle: $\Delta U = 0$.

STUDY METHOD: cover the right side / formula boxes $\rightarrow$ recall from memory $\rightarrow$ check $\rightarrow$ repeat.